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EXAMINING PSYCHOLOGICAL NEEDS AS PREDICTORS OF EXERCISE MOTIVATION AND INTRINSIC MOTIVATION IN COLLEGE STUDENTS

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Abstract – The purpose of this study was to examine whether psychological needs for fulfillment or happiness predicted college students' exercise motivation and intrinsic motivation. A total of $N = 100$ undergraduate college students were recruited to complete self-report measures of exercise motivation, intrinsic motivation, need for psychological fulfillment, and need for happiness. We also measured participants' current levels of psychological fulfillment and happiness. We tested the preregistered hypotheses that both need for fulfillment and need for happiness would be significantly and positively associated with both exercise motivation and intrinsic motivation. We also hypothesized that, in these models including both needs as predictors, need for happiness would be a stronger predictor of exercise motivation, whereas need for fulfillment would be a stronger predictor of intrinsic motivation. Results showed that needs for fulfillment and happiness were not significantly associated with exercise motivation, contrary to our hypotheses. The overall model predicting intrinsic motivation from needs for fulfillment and happiness was statistically significant, however, although neither predictor was uniquely associated with intrinsic motivation. In exploratory analyses, we examined current psychological fulfillment and current happiness as predictors of exercise motivation and intrinsic motivation and found that current fulfillment predicted intrinsic motivation, which is consistent with motivation research more broadly. The pattern of findings observed suggests that needs for fulfillment and happiness may not be critical for exercise motivation, highlighting the importance of more research to examine other psychological needs that may play a larger role in motivating college students to exercise.

Keywords: exercise motivation; intrinsic movitation; psychological needs; happiness; health

Psychological models and theories of wellness emphasize that health and well-being follow from optimal functioning of both the body and mind (e.g., Ryan & Deci, 2000b, 2001). Unfortunately, college students experience myriad stressors (e.g., Dill & Henley, 1998; Heikka et al., 2023; Robotham, 2008), which can result in unhealthy behaviors, subsequently negatively impacting physical and mental health (e.g., O'Connor et al., 2021; Shankar & Park, 2016). For example, college students can become overwhelmed by the pressures of school, resulting in deleterious health behaviors (e.g., Fruehwirth et al., 2023), including poor sleep (e.g., Valerio et al., 2016; Wang & Bíró, 2021), unhealthy eating (e.g., Araiza & Lobel, 2018; Hill et al., 2022), and alcohol use (e.g., Brenman et al., 2024; Park et al., 2004). These behaviors also may have downstream consequences, such as overall poor health (e.g., Kennett et al., 2021; Ribeiro et al., 2018). Attempting to cope with the pressures of college also may result in students gaining weight due to excessive eating related to stress, depression, poor body image, and sleep disorders (Phillips et al., 2016). To combat stress, college students can focus on engaging in healthy behaviors. Exercise, for example, is one of the best available tools to reduce stress and to improve and maintain overall physical and psychological health (e.g., Herbert et al., 2020; Huang et al., 2023; Ruegsegger & Booth, 2018; Warburton & Bredin, 2017; Zhou, et al., 2023). However, people do not exercise enough and attempts to increase exercise engagement often are unsuccessful (e.g., Thivel et al., 2018; Marteau et al., 2012). Because of how beneficial exercise is for health, it is important to understand what motivates college students to exercise. Accordingly, in the present study, we examined psychological needs as potential predictors of college students' motivation for exercise, as well as for their general intrinsic motivation.

Motivation refers to one's reason for engaging in an activity to make effective choices and actions to achieve goals (e.g., Higgins, 2012; Schunk & DiBenedetto, 2021). "To be motivated means *to be moved* to do something" (Ryan & Deci, 2000a, p. 54). Motivated individuals are determined or persistent when attempting to accomplish or gain some outcome. Motivation can be categorized as extrinsic and intrinsic. Extrinsic motivation drives a person to act through external forces or outcomes (e.g., money, a good grade, social status) that are independent of one's own internal desires and interests (e.g., Morris et al., 2022; Ryan & Deci, 2020). Intrinsic motivation, on the other hand, is an internal feeling or desire that drives a person to act (Ryan & Deci, 2020). Individuals who are intrinsically motivated engage in those actions because of interest or internal enjoyment in the absence of external rewards (e.g., Kruglanski et al., 2018; Morris et al., 2022; Ryan & Deci, 2020). In other words, intrinsic motivation is an internally driven pursuit to accomplish something that has personal significance to an individual.

Intrinsic motivation relates to satisfaction of basic psychological needs that people are personally trying to fulfill and that promote psychological growth (e.g., Legault et al., 2017; Ryan & Deci, 2020; Ryan et al., 2019; Vansteenkiste & Ryan, 2013; Vansteenkiste et al., 2023; Warburton et al., 2020). Because of these features, intrinsic motivation is highly useful for promoting maintenance of goal-related behaviors that require motivation and sustained effort over time. According to Basic Psychological Needs Theory (BPNT), a sub theory of the self-determination theory of motivation, satisfaction of one's basic psychological needs for autonomy (i.e., the ability or right to govern oneself and one's activities), competence (i.e., mastery or self-efficacy over one's activities), and relatedness (i.e., the sense of belonging or connection to other people) promote maximum goal-related persistence and performance (e.g., Deci & Ryan, 2000; Ryan & Deci, 2000b; Vansteenkiste et al., 2020). Such persistence and performance are indicative of high levels of motivation and engagement related to a variety of tasks and activities (e.g., Ryan & Deci, 2000b; Vansteenkiste et al., 2020). Previous research supports this, showing that intrinsic motivation is closely linked to feeling fulfilled in life, as indicated by such psychological need satisfaction (e.g., Deci & Ryan, 2000; Ryan & Deci, 2000b; Vansteenkiste & Ryan, 2013). Self-determination theory also posits that peak psychological health and well-being stem from fulfillment of these basic psychological needs. Specifically, satisfaction of needs leads to further motivation and positive psychological health outcomes, whereas a lack of satisfaction of basic psychological needs (i.e., not meeting one's needs) is linked to low motivation and poor health and well-being (e.g., Martela et al., 2023; Ryan & Deci, 2000b; Ryan & Deci, 2017).

Among college students, motivation and achievement tend to be negatively associated with unhealthy behaviors (e.g., Allen et al., 2022; El Ansari et al., 2013; Hershner, 2020; Petrucci et al., 2020). This is concerning because motivation can be especially important for health behaviors such as exercise, which, as noted earlier, requires sustained effort and is an important tool for reducing stress and optimizing health. Fortunately, motivation related to physical health and physical activity itself also can be psychologically satisfying for people (e.g., Ipolyi et al., 2021; Xu et al., 2021). To increase motivation for exercise, then, it may be helpful to understand what drives that motivation. Some people might be motivated to exercise by a need to feel fulfilled in life. For example, studies have shown that fulfilled psychological needs such as autonomy are associated with motivation for physical activity (e.g., Cox et al., 2008; Schneider & Kwan, 2013), greater intentions to participate in physical activity in the future (e.g., Sánchez-Oliva et al., 2020; Shen et al., 2007), and physical activity behavior (e.g., Hagger et al., 2003; Kang et al., 2020). However, if a person's need for fulfillment is not yet met, but rather is desired, that desire also may drive motivation. Accordingly, it is possible that a person's need for psychological fulfillment (e.g., how *important* it is to them that they are free to decide how to live their life, versus whether they feel they *are* free to do so) relates to their exercise and/or intrinsic motivations, as well.

Research also has shown that happiness is associated with intrinsic motivation (e.g., Patrick et al., 2007). Happiness is perceived as a mental state (e.g., Jongbloed & Andres, 2015; Raibley, 2012) and conceptualized by most people as positive emotions or life satisfaction, with the absence of negative emotions as a third commonly noted component (Argyle, 2001). People may feel happy when they attain either their intrinsic or extrinsic aspirations (Kasser & Ryan, 1996) or when they pursue or engage in acts that align with their goals (e.g., Sheldon et al., 2010; Wang & Milyavskaya, 2020). Studies have shown that positive affect predicts intrinsic motivation and motivation to devote additional time and effort to attaining goals (e.g., Haase et al., 2012; Løvoll et al., 2017). Aside from these benefits of *current* happiness for motivation, people might be motivated generally and/or to exercise by a desire, or *need*, to feel happy (e.g., how *important* it is to them that they consider themselves a happy person, versus whether they

do consider themselves happy). This is especially possible given popular knowledge of the empirically supported relationship between exercise and happiness (e.g., Richards et al., 2015; Zhang & Chen, 2019).

There is limited research, however, examining whether psychological *needs for* fulfillment (i.e., autonomy, competence, and relatedness) or happiness (i.e., positive emotion and life satisfaction) might relate to intrinsic and/or exercise motivations. Researchers have expanded on self-determination theory to propose a distinction between psychological needs themselves serving as motivation and psychological needs existing as requirements to be satisfied (e.g., McClelland, 1985; Sheldon, 2011). According to this two-process model, met (i.e., satisfied) needs differ from not-yet-met (i.e., dissatisfied) needs in that they reflect different timepoints in a motivated action sequence (Sheldon, 2011). Need satisfaction and need dissatisfaction are thought to be related but conceptually distinct from each other, and it has been suggested that an unmet need should be motivating (see Neubauer et al., 2018 for a review). However, findings on the motivational effects of unmet needs are mixed (Neubauer et al., 2018). Therefore, more research is needed to understand how desired, or unmet, needs relate to intrinsic motivation and whether they also may relate to exercise motivation, specifically.

The Present Study

As outlined above, intrinsic motivation is closely related to fulfilled psychological needs and successful goal attainment, and understanding both unmet and met needs is important for understanding motivation. In the present study, we investigated whether a person's unmet but desired needs for fulfillment and happiness relate specifically to exercise motivation and to general intrinsic motivation. Our key interest was in examining whether a need for fulfillment and/or a need for happiness predicted college students' exercise motivation, given that exercise may serve as a healthy way for college students to mitigate the negative health effects of stress. Additionally, because college students regularly engage in a wide range of academic and personal tasks that require motivation, we tested whether either need was associated with intrinsic motivation in our sample of college students. We hypothesized that need for fulfillment and need for happiness would be positively associated with exercise motivation, but that need for happiness would be a stronger predictor of exercise motivation. We also predicted that need for fulfillment and need for happiness would be positively associated with intrinsic motivation, but that need for fulfillment would be a stronger predictor of intrinsic motivation. Prior to data collection, the study hypotheses and analyses plan were preregistered via the Open Science Framework (osf.io/2n95c).

Method

Participants

A total of N = 100 undergraduate students were recruited from the psychology subject pool at Angelo State University to participate in this research online in exchange for course credit. A power analysis was conducted using G*Power 3.1.9.6 for a multiple regression analysis with two predictor variables. This yielded a target sample size of 100 to detect a small-to-moderate effect size of f^2 = .10 at alpha .05 with .80 power. The study was available on Sona Systems during a single semester. Participants signed up for the study through their Sona Systems account and completed the study via Qualtrics survey software. For their involvement, participants received Sona credit to be applied toward requirements and/or extra credit opportunities in their psychology courses.

Participants were primarily female (Sex: 85 Female, 14 Male, 1 declined to respond) and ranged in age from 18 to 52 years old (M = 20.27, SD = 4.84). Most of the study participants were between the ages of 18 and 25, but three participants reported ages above 25 (i.e., 36, 48, 52). Although these participants are older than traditional college students, it is not unusual for us to have a few older participants in some samples at our university. As such, we retained their data. Importantly, whether including or excluding two age outliers (i.e., 48 and 52 years old, which are greater than 3.5 standard deviations above the mean age), the pattern of results across the analyses are the same. Reported race/ethnicity of the sample was 42% European American/White, 32% Hispanic/Latino American, 15% more than one race, 4% African American, 3% Other, 2% Asian American, and 1% Native American/American Indian (1 declined to respond). There were no statistically significant sex or race/ethnicity differences in any of the study variables of interest, and age was not significantly correlated with any of the study variables. Further details regarding participant characteristics are presented in Table 1. All protocols and procedures of this study were approved by the human-subjects review board at Angelo State University.

Procedure

This study was conducted online in one session. The survey informed participants that they were participating in a study that would assess student

Table 1
Full report of sociodemographic characteristics of the study participants.

Sample Characteristic	*n*	%	Range	*M*	*SD*
Sex					
Female	85	85.9%			
Male	14	14.1%			
Declined to Respond	1				
Gender					
Woman	82	82.8%			
Man	13	13.1%			
Non-binary/Third gender	2	2.0%			
Genderqueer/Gender non-conforming	2	2.0%			
Something else	0	0.0%			
Prefer not to say	0	0.0%			
Declined to Respond	1				
Race/Ethnicity					
African American	4	4.0%			
Asian American	2	2.0%			
Hispanic/Latino American	32	32.0%			
Native American/American Indian	1	1.0%			
European American/White	42	42.0%			
Other	3	3.0%			
More than one race/ethnicity selected:					
African American and Hispanic	3	3.0%			
African American and White	1	1.0%			
Asian American and White	2	2.0%			
Hispanic and White	9	9.0%			
Declined to Respond	1				
Age	99		18-52	20.27	4.84
Declined to Respond	1				
Body Mass Index	97		16.46-55.91	26.10	7.46
Declined to Respond	3				

Note: Total sample size = 100

motivation and health. Participants completed a consent form and then completed the study questionnaires presented in randomized order. Upon completion of the survey, participants were provided with an online debriefing statement and thanked for their participation.

Measures

The two predictors of interest were "*need for* fulfillment" and "*need for* happiness." To measure these needs, we modified existing questionnaires of *current* psychological fulfillment and *current* subjective happiness to reflect how important these needs were to participants. These measures and their necessary modifications are described below.

Need for Fulfillment

The 21-item Basic Psychological Needs Satisfaction Scale (Ryan & Deci, 2000b) was used to assess psychological need fulfillment. This scale measures the extent to which basic psychological needs for autonomy, competence, and relatedness are met. Here, however, we were interested in measuring *need for* fulfillment, therefore we modified the items. For example, the original item "I feel like I am free to decide for myself how to live my life" was changed to "[It's important to me that] I feel like I am free to decide for myself how to live my life;" another example modified item is "[It's important to me that] people are generally pretty friendly towards me." Participants rated the extent to which each statement was true for them on a scale from 1 (*not at all true*) to 7 (*very true*). The items were averaged to create an overall score, with higher values reflecting a greater need for fulfillment. Participants also completed the original measure to assess *current* psychological need fulfillment. Both versions showed very good reliability. Need for fulfillment: Range = 1 to 7, M = 5.03, SD = .76, α = .84; Current fulfillment: Range = 1 to 7, M = 4.81, SD = .83, α = .87.

Need for Happiness

The 4-item Subjective Happiness Scale was used to assess happiness (Lyubomirsky & Tucker, 1998). Here, we were interested in measuring one's *need for* happiness, therefore we modified the items. For example, the original item "In general, I consider myself:" rated on a scale from 1 (*not a very happy person*) to 7 (*a very happy person*) was changed to "In general, [it's important to me to] consider myself:" rated from 1 (*not a very happy person*) to 7 (*a very happy person*). Another example is "Compared to most of my peers, [it's important to me to] consider myself:" rated from 1 (*less happy*) to 7 (*more happy*). Each item was assessed using its own response scale. Scores on all items were summed, with higher values indicating a greater need for happiness. Participants also completed the original scale to measure *current* happiness levels. Reliability was acceptable for the modified version and adequate for the

original version (Taber, 2018). Need for happiness: Range = 4 to 28, M = 17.81, SD = 3.89, α = .60; Current happiness: Range = 4 to 28, M = 16.70, SD = 4.80, α = .77.

Exercise Motivation

The Behavioral Regulation of Exercise Questionnaire 3 was used to assess exercise motivation (Markland & Tobin, 2004; Wilson et al., 2006). We included 20 items to measure motivation related to exercise. Some example items include "It's important to me to exercise regularly" and "I enjoy my exercise sessions." Participants rated each item on a scale from 0 (*not true for me*) to 4 (*very true for me*). The measure assesses several types of motivation, but we averaged all items into a single score reflecting overall motivation for exercise. This 20-item version showed very good reliability; Range = 0 to 4, M = 2.48, SD = .65, α = .87.

Intrinsic Motivation

The achievement motivation items of the Work and Family Orientation Questionnaire (Spence & Helmreich, 1983) were used to assess three facets of intrinsic motivation (i.e., work motivation/achievement, mastery, and competitiveness). Due to researcher error, the competitiveness subscale was omitted and only the 14 items reflecting the work motivation/achievement subscale and the mastery subscale were included in this study. The work motivation/achievement factor represents "the desire to work hard and to do a good job" and the mastery factor reflects "a preference for difficult, challenging tasks and for meeting internally prescribed standards of performance excellence" (Spence & Helmreich, 1983). Participants rated items such as "I like to work hard" and "Once I undertake a task, I persist" on a scale from 1 (*strongly disagree*) to 5 (*strongly agree*). A total score was calculated by summing all work motivation/ achievement and mastery items. The scale showed adequate reliability, with higher scores reflecting more intrinsic motivation; Range = 14 to 70, M = 52.28, SD = 6.81, α = .78.

Demographics

Participants reported their sex, gender identity, age, height, weight, and race/ethnicity.

Results

Data were analyzed using Pearson product-moment correlations and simultaneous multiple regression analyses. Correlations for all study variables are presented in Table 2. Full regression results for hypothesis tests are presented in Table 3.

Exercise Motivation

Correlational analyses showed that neither need for fulfillment (r = .11, p = .274) nor need for happiness (r = -.03, p = .797) were statistically significantly correlated with exercise motivation, contrary to our hypotheses.

Preregistered Analysis

To test our predictive hypotheses related to exercise motivation, we entered need for fulfillment and need for happiness into a simultaneous multiple regression model predicting exercise motivation. No significant associations were observed between either predictor variable of interest and exercise motivation. The overall model was not statistically significant, R^2 = .02, $F(2, 97)$ = .72, p = .490. Need for fulfillment (β = .12,

Table 2

Correlations among all study variables, both primary and exploratory.

	1	2	3	4	5
1. Need for Fulfillment					
2. Need for Happiness	.20*				
3. Current Fulfillment	.46***	.50***			
4. Current Happiness	.09	.76***	.59***		
5. Exercise Motivation	.11	-.03	.13	.12	
6. Intrinsic Motivation	.19	.22*	.37***	.34**	.11

Note: Pearson product-moment correlation coefficients presented. *p < .05, **p < .01, ***p < .001

Table 3

Full regression output for primary hypothesis tests predicting exercise motivation and intrinsic motivation from need for fulfillment and need for happiness.

Dependent Variable:	Exercise Motivation			Intrinsic Motivation		
Overall Model:	R^2 = .02 $F(2, 97)$ = .72, p = .490			R^2 = .07 $F(2, 97)$ = 3.64, p = .030		
Predictor Variables:	***β***	***SE***	***p***	***β***	***SE***	***p***
Need for Fulfillment	.12	.09	.245	.16	.90	.119
Need for Happiness	-.05	.02	.628	.18	.18	.069

SE = .09, p = .245) and need for happiness (β = -.05, SE = .02, p = .628) were not significantly associated with exercise motivation, contrary to our hypotheses.

Exploratory Analysis

Next, we explored associations between current fulfillment and current happiness with exercise motivation. The overall model predicting exercise motivation from current fulfillment and current happiness also was not statistically significant, R^2 = .02, F(2, 97) = .98, p = .380. Neither fulfillment (β = .10, SE = .10, p = .437) nor happiness (β = .06, SE = .02, p = .631) predicted exercise motivation when considered uniquely.

Results did not support any associations of needs for fulfillment or happiness nor current fulfillment or happiness with exercise motivation.

General Intrinsic Motivation

Contrary to expectations, need for fulfillment was not statistically significantly correlated with intrinsic motivation (r = .19, p = .054). However, need for happiness was significantly, positively correlated with intrinsic motivation (r = .22, p = .030), supporting our hypothesis that these variables would be positively associated.

Preregistered Analysis

To test our hypotheses related to intrinsic motivation, we entered need for fulfillment and need for happiness into a simultaneous multiple regression model predicting intrinsic motivation. The overall model was statistically significant, R^2 = .07, F(2, 97) = 3.64, p = .030, providing some support for our hypothesis that these predictors combined would account for variance in intrinsic motivation. However, neither variable was uniquely associated with intrinsic motivation in this model; specifically, need for fulfillment (β = .16, SE = .90, p = .119) and need for happiness (β = .18, SE = .18, p = .069) both were non-significant predictors of intrinsic motivation in the context of one another. This result does not support our additional hypothesis that need for fulfillment would be more strongly associated with intrinsic motivation.

Exploratory Analysis

Next, we explored associations of current fulfillment and current happiness with intrinsic motivation. The overall model was statistically significant, R^2 = .16, F(2, 97) = 9.16, p < .001. In this model, current fulfillment was significantly positively associated with intrinsic motivation (β = .26, SE = .94, p = .025), indicating that intrinsic motivation is higher among participants who report higher levels of current need fulfillment. However, the unique association of current happiness with general intrinsic motivation was not significant (β = .18, SE = .16, p = .112).

Discussion

In a preregistered study, we examined whether college students' exercise motivation and intrinsic motivation were driven by a psychological need for fulfillment or a need for happiness in life. We hypothesized that need for fulfillment and need for happiness would be positively associated with exercise motivation, and that need for happiness would be a stronger predictor of exercise motivation. We also predicted that need for fulfillment and need for happiness would be positively associated with intrinsic motivation, and that need for fulfillment would be a stronger predictor of intrinsic motivation. Results showed that needs for psychological fulfillment and happiness may not be important factors for predicting exercise motivation or for uniquely predicting intrinsic motivation. More specifically, none of the needs variables were statistically significantly associated with exercise motivation. Although need for happiness was positively associated with intrinsic motivation, as hypothesized, this association did not remain statistically significant when considered simultaneously with need for fulfillment. However, needs for fulfillment and happiness together did account for significant variance in intrinsic motivation. Finally, the exploratory analyses showed that current fulfillment uniquely predicted intrinsic motivation in the model considering it together with current happiness. This finding is consistent with previous literature on intrinsic motivation and provides further support for current psychological need fulfillment as a key predictor of intrinsic motivation.

Neither need for fulfillment nor need for happiness were associated with exercise motivation in this study. These null findings suggest many interesting possibilities for other variables not included here that may be more informative for understanding exercise motivation. For example, research has shown that exercise motives are related to body image (e.g., Panão & Carraça, 2020; Teixeira et al., 2012; Tylka & Homan, 2015). Considering this, it is possible that aesthetic needs may be more useful than needs for fulfillment or happiness for predicting exercise motivation. It also has been suggested that the rejection and discrimination people higher in body weight experience may prevent them from fulfilling the fundamental need for belonging (Magallares et al., 2011). Research does support this notion, as people higher in body weight have lower quality social relationships (e.g., Albano et al., 2019; Carr & Friedman, 2006). Relatedly, individuals whose social network accepts, rather than rejects, their weight tend to

show more positive psychological and physical health outcomes (e.g., Ata et al., 2007; Logel et al., 2014). Thus, another potential predictor of exercise motivation may be a need for belonging. Specifically, individuals who value their social life may be more inclined to exercise to achieve a desired weight/figure that affords them acceptance from others, which may help them meet their psychological need to belong.

Another potential important need related to exercise motivation may be the need to protect or bolster self-esteem to avoid social exclusion (Leary et al., 1995). For example, individuals concerned about weight-based rejection may be motivated to exercise to protect their self-esteem from threats of ostracism. Additionally, those who equate worth to their body weight or figure may be more motivated to exercise in the service of body- or self-esteem (e.g., Sani et al., 2016; Wilson & Rogers, 2002; Zervou et al., 2017). Each of these variables reflects an interesting target for future research, building on the knowledge gained from the present work that needs for fulfillment and happiness may not relate to exercise motivation.

Our tests of current fulfillment and current happiness predicting exercise motivation were exploratory and the lack of a significant association between current fulfillment and exercise motivation is inconsistent with previous research on psychological need fulfillment and exercise. For example, overall psychological need fulfillment (i.e., all three facets of the Basic Psychological Needs Satisfaction Scale: autonomy, competence, and relatedness) and the separate autonomy and relatedness facets of psychological need fulfillment have been associated with motivation for physical activity (Cox et al., 2008). However, Cox et al. (2008) studied adolescents and the present work utilized adult college students, for whom other needs (e.g., physical thriving, pleasure) or life circumstances (e.g., sports training, dating) may be more salient in relation to exercise motivation. Other researchers have observed positive associations between psychological need satisfaction and exercise motivation among college students but have noted that various scoring protocols for the exercise motivation questionnaire used here may explain differences in findings and interpretations across studies that use the measure (Chu & Zhang, 2022). It also is important to note that the non-significant observed predictor variables' associations with exercise motivation were small and may not have been detectable with the a priori powered sample size, which we calculated using a more moderate effect size based on effect sizes in previous studies relating psychological need satisfaction to exercise (e.g., Chu & Zhang, 2022; Cox et al., 2008). Regardless of statistical significance, however, the associations observed in our sample were much smaller than in previous studies. This highlights a need for further research into psychological needs for fulfillment and happiness as they relate to exercise motivation among college students both similar to and different from those in the present sample.

Although the hypotheses related to exercise motivation were not supported, the exploratory finding that current psychological fulfillment was a significant predictor of intrinsic motivation, even in the context of current happiness, is consistent with previous research showing that when an individuals' needs are fulfilled, they are more motivated (e.g., Ryan & Deci, 2000b; Vansteenkiste & Ryan, 2013). This may be because those whose needs are fulfilled do not have to expend as much energy satisfying needs and instead can focus on motivating themselves. This predictive utility of current psychological need fulfillment for intrinsic motivation observed here is consistent with BPNT and unsurprising based on a wealth of motivation literature. As such, this adds validity to the overall contributions of this work.

Limitations and Future Directions

One limitation of the present study is that we modified the original measures of psychological need fulfillment and subjective happiness to assess *needs for* fulfillment and happiness. The happiness measure, in particular, may have been confusing for participants following modification, given that the items and response scales of the original measure have different wording and may not clearly represent opposite ends of a continuum once modified. For example, responding to the item "In general, it is important to me to consider myself: *not a happy person* versus *a very happy person*." may not be the same as responding to the original item (i.e., "In general, I consider myself: *not a happy person* versus *a very happy person*."). Furthermore, the endpoints of the response scale for such an item may not reflect the same construct. Although reliability coefficients for the modified fulfillment and happiness measures were acceptable, it is possible that the items and/or response scales may have been confusing for participants or may require further/other modifications to ensure they are measuring the same construct. As such, future research should use other measures of these needs or conduct preliminary studies to assess clarity and accuracy of the items as modified in this study.

Also related to our measures, need for fulfillment was negatively skewed in the present data. There was still variability in the scores on need for fulfillment, but its skew could be a potential explanation for its unexpected non-significant association with general intrinsic

motivation here, which is inconsistent with the literature on current need fulfillment and intrinsic motivation. Additionally, the competitiveness subscale of the intrinsic motivation measure was omitted from the survey administered to participants, due to researcher error. Although this concern is alleviated somewhat because the remaining items for each scale were adequately reliable, future research should replicate this study including the competitiveness subscale of the intrinsic motivation measure. Future studies also should include other measures of exercise motivation, given the varied scoring procedures for the exercise questionnaire used in this study.

Another limitation is that this study was conducted using a general sample of undergraduate college students. However, exercise motivation may be related to needs for fulfillment and happiness in life in more targeted samples, such as student athletes. Therefore, future research could examine whether needs for fulfillment and/or happiness are important predictors of exercise motivation among athletes or students who regularly engage in exercise. Relatedly, we also did not measure participants' current exercise behavior, which may be important. Although our focus was on psychological needs and their relation to motivation (rather than behavior), future research extending this work to exercise behavior or controlling for participants' current levels of exercise engagement will be informative. Experimental or longitudinal studies also may help determine directionality and the extent to which current exercise patterns lead to or stem from exercise motivation and/or relate to psychological needs.

Given the demographic characteristics of our sample, results from this work also are limited in their generalizability. Our university and the present sample comprise a greater number of students who identify as Hispanic and fewer students who identify as other races or ethnicities, compared to college students and samples nationally. This may have important implications for our findings, as both psychological needs and motivation may differ for individuals of different races/ethnicities or cultural backgrounds. Race differences were not observed in the present study across our needs and motivation variables, which is consistent with related theorizing and research on the cross-cultural applicability of self-determination theory (e.g., Ryan & Deci, 2020). However, other theorizing and research do support the existence of cultural differences in such variables (e.g., Nolen, 2020), meaning further investigation is warranted. Our sample also consisted of primarily self-identified women, meaning that the present results may not generalize beyond women. As such, future studies examining how psychological needs relate to exercise motivation among college students of different races and genders will be important for understanding the applicability of the present findings to wider college-student populations and for illuminating ways to motivate college students more broadly. Finally, studies testing whether associations between psychological need fulfillment and exercise motivation could be moderated by race, age, sex, or gender, and why, also could be interesting next steps for this research. For example, it may be useful to further examine the potential for sex or gender differences in associations of various psychological needs with exercise motivation, given previous research showing that males and females are motivated to exercise by different factors (e.g., strength versus appearance, respectively; Egli et al., 2011).

Implications and Conclusions

The broad aim of this study was to investigate possible factors related to exercise motivation and intrinsic motivation among college students. We aimed to understand whether needs for fulfillment or happiness in life predict students' motivations for exercise and/or their motivation in general. Our findings align with previous research demonstrating a consistent association between psychological fulfillment and intrinsic motivation while potentially ruling out needs for fulfillment and happiness as predictors of exercise motivation. These results have implications for college students, as well as professionals who work with them. For example, this work provides further evidence for the importance of basic psychological need fulfillment for general intrinsic motivation, which is critical for college students' success. Counselors, coaches, faculty, and other professionals working directly with college students should aim to foster autonomy, competence, and relatedness in the tasks and activities they assign (e.g., offer agency over schedules and activities, encourage mastery in sports and the classroom, provide opportunities for engagement and teamwork to build connections with their peers).

Importantly, the lack of associations of either need for fulfillment or need for happiness with exercise motivation have implications for our understanding of what motivates college students to exercise, particularly by highlighting the need for additional research into this important question. These findings are particularly applicable to research professionals. Specifically, it will be important to identify other personal needs that may be more relevant to exercise motivation, which in turn can inform academic professionals more directly on how to motivate exercise behavior to promote optimal physical and psychological health among college students. Determining predictors of exercise motivation in college

students has the potential to help them remain active and healthy during their college years and beyond.

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Author Note

The preregistered study plan and hypotheses, a de-identified participant data file, a data codebook, and analytical code/syntax can be found on the Open Science Framework at osf.io/kxesa. The authors report there are no competing interests to declare.

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